

Suma Pranitha Konduru

(716)-232-0669 | [Email](#) | [LinkedIn](#)

SUMMARY

Data driven Software Engineer with 2+ years of experience in ML, NLP, and Distributed Systems. Skilled in Python, SQL, PyTorch, Spark, and cloud platforms with expertise in ETL pipelines, LLM fine-tuning, and scalable data workflows. Passionate about applying data-driven approaches to solve business challenges, optimize pipelines, and enable predictive analytics in production environments.

EDUCATION

The State University of New York - Master of Science in Computer Science (AI/ML)

Jan 2024 – May 2025

WORK EXPERIENCE

Research Assistant – University at Buffalo

May 2025 - present

- Designed scalable data labeling, ETL, and data validation pipelines to support NLP, LLM training, and model fine-tuning experiments.
- Automated data preprocessing and annotation workflows using Python, Airflow, and SQL, ensuring higher throughput and reduced manual effort.
- Authored reproducibility guides, dataset documentation, and ML experiment tracking frameworks (MLflow) for academic and research teams.
- Collaborated with researchers to optimize model deployment on AWS balancing latency vs. compute cost trade-offs.

Software Engineer - Contriver

Jul 2023 - Dec 2023

- Built a REST APIs and microservices for structured data ingestion and transformation, enabling real-time analytics and labeling pipelines across distributed systems.
- Implemented ETL and streaming pipeline optimizations using Kafka and Spark, reducing data retrieval latency by 20% and cutting operational costs.
- Created interactive dashboards in Power BI and Tableau to visualize KPIs, labeling quality, and forecasting trends for business stakeholders.
- Improved scalability and fault tolerance of ML pipelines by containerizing workflows with Docker & Kubernetes, ensuring high availability.
- Utilized Excel, SQL, and Python to analyze large datasets and create dashboards supporting data-driven decisions

Software Engineer Intern - Contriver

Jan 2023 - Jun 2023

- Developed a collaborative code editor with distributed computing modules for real-time synchronization, version control, and scalable ML workflows.
- Applied object-oriented programming (Java, Python) and modular design to improve maintainability of large-scale data engineering projects.
- Diagnosed root causes of latency and bottlenecks in distributed pipelines, improving throughput and reducing inference delays by 25%.
- Supported A/B testing of new ETL pipeline strategies, providing insights that enhanced data labeling accuracy.

PROJECT EXPERIENCE

License Plate Detection with privacy enhancement.

Jan 2025 - May 2025

- Designed a real-time license plate recognition system integrated with a distributed ETL pipeline, achieving >90% accuracy.
- Optimized CPU/GPU scheduling and fault-tolerant orchestration, reducing latency and false positives (<5%)

American Sign Language Detection using Deep Neural Networks

Aug 2024 - Dec 2024

- Designed an ASL-to-text recognition system with >95% accuracy and 40% lower inference latency.
- Deployed on AWS for scalable real-time inference, improving accessibility for live users.

SKILLS

Programming & Data Processing: Python, SQL, R, Java, JavaScript, PyTorch, TensorFlow, Scikit-learn, PySpark

AI & ML: LLMs, NLP, DL (CNN, RNN), ML, Computer Vision

MLOps & Automation: MLflow, Apache Airflow, CI/CD (Jenkins, GitHub), Docker, Kubernetes, REST APIs, ETL Pipelines

Cloud & Big Data: AWS (S3, Lambda, Redshift), GCP (BigQuery, Dataflow), Hadoop, Spark, Kafka, Snowflake

Data Visualization & BI: Tableau, Power BI, Matplotlib, Excel

Databases & Storage: MySQL, PostgreSQL, MongoDB, NoSQL, Data Warehousing, Data Lakes

CERTIFICATIONS

AWS Certified Developer – [Associate](#)

HP LIFE: [AI for Beginners](#)